BMCS 4480
Fall 2026

Assignment 1

Submission Due: **Sep 30th, 11:59pm**

- *Submit your work directly to Courseworks. For Q1-3, include your Python script with output figures and responses (zip file), or Python Jupyter notebook (pdf). Alternatively submit your R/MATLAB code and writeup containing figures (zip).*
- *Existing libraries such as Scanpy, Seurat, and Scikit-Learn can be used, but we encourage implementing your own functions. Please mention the names of all collaborators you may have discussed these problems with.*
- *Usage of AI is only permitted for debugging and searching references for library documentation - a print out of your AI tool chat/cowork session should be attached.*

Question 1. Download preprocessed single-cell gene expression data for ~4K PBMCs (peripheral blood cells) from a healthy donor from here. Note that the top 3 lines of the mtx file are header lines. The third line contains the total number of rows in all the three files in this folder (genes.tsv, barcodes.tsv, matrix.mtx). The next lines (starting from line 4) contain indices for gene id, cell id, and expression counts (more details here).

Q1a. Load the above data and construct a dense matrix of genes by cells from the sparse list format. What is the size (shape) of this matrix, how many cells and genes are detected? [5 pts]

Q1b. Plot a histogram of log10 of library size (i.e. total counts per cell). What type of probability distribution can be a good fit? Do you think the data requires filtering of cell barcodes? If yes, propose an explicit filtering rule based only on the observed data and justify your choice. Do **not** use a threshold taken from Scanpy/Seurat tutorials. Report how many cells your rule removes.
Plot the histogram after filtering. Compute the median of library size across all cells after any filtering. Repeat the downstream calculation with a threshold 20% more and 20% less stringent. How much does the median library size change? Based on this sensitivity analysis, was your original filtering decision consequential? [15 pts]

Q1c. Plot a histogram of log10 of the total number cells that each gene is expressed in. Is this distribution unimodal or multimodal (point to evidence from the data)? If the latter, how many genes remain if you filter the lower mode as noisy genes [5 pts]?

Question 2. Normalize and visualize the data as follows:

Q2a. Normalize the data in Q1 (post filtering) with global scaling to the median library size [10 pts].

Q2b. Embed the normalized data to 2D using t-SNE or UMAP or another dimensionality reduction method. Compare your results with the first two principal components. Discuss the assumptions of your method of choice [10 pts].

Q2c. Color cells in the embedded 2D map by log10 of library size. Interpret the plot [5 pts].

Q2d. Color cells in the 2D map by normalized expression of major immune cell markers. Which cell types can you detect? [10 pts]

Question 3. Calculate the correlation between library size and expression for every highly variable gene (after scaling to median library size performed in Q2a). Plot the distribution. Fit a linear regression model to expression data as a function of library size. Then use the model to remove ("regress out") any remaining effect of library size as a technical artifact and redo Q2b-c. Explain if/why you used a regularizer . Limit this analysis to highly variable genes. [20 pts]

For the following questions, please show all necessary work for full points.

Question 4. Solve problem 2.13 of PRML textbook [20 pts]. *Note: This is multivariate!*

Question 5 [Bonus]. Solve problem 3.12 of PRML textbook [Bonus 10 pts]. *Hint: Use Bayes' rule.*